Einstein Defect at the Orbifold Boundary: Multiple-Core Obstructions and Actual Subthreshold Degeneration

Research continuation

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Abstract

We continue the smooth-attainment question for the four-dimensional functional $P(g) = \int |\operatorname{Ric}^\circ_g|^2$. Actual monopole supply transfers to every metric on a fixed orbifold obtained by replacing finitely many Ricci-flat ALE cores. The entire supplied-class span has anti-self-dual restrictions on each core; this preserves the convex-hull budget and yields an orientation-sensitive curvature obstruction. Conversely, ordinary-orientation rational balls really embed in every single blow-up. Combining a classical Godeaux degeneration with the project's negative Kähler–Einstein Burns insertion gives a fixed non-Einstein closed manifold with bubbling sequences strictly below $8\pi^2$. This separates a strict upper bound from smooth attainment. No global minimum or resolution of the four outstanding questions is asserted.

1 Definitions and inputs

All manifolds are oriented. Curvature norms are fixed by

$$8\pi^2\chi(M) = \int_M \left(|W^+|^2 + |W^-|^2 + \frac{s^2}{24} - \frac{|\operatorname{Ric}^\circ|^2}{2} \right), \tag{1}$$

$$12\pi^2\tau(M) = \int_M (|W^+|^2 - |W^-|^2). \tag{2}$$

Write $T(M) = 2\chi(M)+3\tau(M)$ and $P_*(M) = \inf_g P(g)$, with the infimum over all smooth metrics. For an orbifold, χ_{curv} and τ_{curv} denote the corresponding curvature integrals. In particular τ_{curv} is *not* necessarily the signature of its real intersection form.

The inherited analytic input for Section 4 is the project's strict Burns insertion on a negative Kähler–Einstein background. Its statement is in the supplied *01_Research_Synthesis.tex*, Section 3, equations (insertion), (phi), and (kernel): in a kernel direction of $\mathsf{R}^+ = W^+ + sI/12$,

$$P(g_\varepsilon) = P(g) + 8\pi^2 - 4\pi^2|T_p|^2\varepsilon^2\log(1/\varepsilon) + O(\varepsilon^2). \tag{3}$$

Here T_p is the trace-free restriction of R^+ to the orthogonal two-plane. On a Kähler–Einstein surface, the eigenvalues of R^+ are $(s/4, 0, 0)$, so $|T_p|^2 = s^2/32 > 0$ when $s < 0$. We use this project result as an input, not as a newly verified literature theorem.

2 Actual monopole supply through several ALE cores

Let $\mathcal{C}$ be a finite collection of determinant classes of fixed spin^c structures on M, each admitting an unperturbed Seiberg–Witten solution for every smooth metric. Suppose compact cores of finitely many oriented Ricci-flat ALE spaces (N_i, g_i) embed disjointly in M. Their ends are S^3/Γ_i, with positive ALE decay order and two differentiated estimates. Replacing the cores by their marked quotient cones gives a compact orbifold O with isolated quotient singularities.

We use compatible orbifold spin^c extensions of the boundary structures; the finite isotropy representations are retained throughout.

Because the links are real homology spheres, there is an orthogonal splitting

$$H^2(M;\mathbb{R}) = H^2_{\mathrm{orb}}(O;\mathbb{R}) \oplus \bigoplus_i H^2(N_i;\mathbb{R}).$$

Write $v = (v_0; v_1, \dots, v_r)$ and set

$$B_O = \max_{v \in \mathrm{Conv}(\mathcal{C})} v_0^2, \qquad D_i = \frac{1}{2\pi^2}\int_{N_i} |W^+|^2.$$

Theorem 1. *Assume $B_O > 0$. The restriction to each N_i of the entire real span of $\mathcal{C}$ is represented by L^2 anti-self-dual harmonic two-forms. Consequently*

$$B_O \geq \max_{v \in \mathrm{Conv}(\mathcal{C})} v^2. \tag{4}$$

Every smooth orbifold metric g on O satisfies

$$\boxed{P_{\mathrm{orb}}(g) \geq 8\pi^2 \left(\frac{2}{3} B_O - T(M) + \sum_i D_i \right).} \tag{5}$$

Proof. Fix an arbitrary orbifold metric g. Near each quotient point its lift has $g = \delta + O(r^2)$ in normal coordinates, with two differentiated bounds. If the ALE order is $\nu > 0$, its rescaling by a^2 has error $O(a^\nu r^{-\nu})$. Choose $r_a = a^q$ with $0 < q \leq \nu/(\nu+2)$ and use a radial cutoff on $r_a < r < 2r_a$. The annular curvature satisfies

$$|\,\mathrm{Rm}\,| \leq C\left[1 + a^\nu r_a^{-\nu-2} + (r_a + a^\nu r_a^{-\nu-1})^2\right] \leq C.$$

The resulting smooth metrics g_a on M have uniformly bounded volume, a uniform lower scalar-curvature bound, and vanishing annular curvature energy. The exact inner metrics are Ricci-flat, hence

$$P(g_a) \longrightarrow P_{\mathrm{orb}}(g). \tag{6}$$

These are direct metric interpolations, not Einstein desingularizations.

Choose actual solutions for every supplied class. The maximum principle bounds the spinors uniformly, hence bounds F_A^+ pointwise. The identity

$$\int_M (|F_A^+|^2 - |F_A^-|^2) = 4\pi^2 a^2$$

then bounds the determinant curvature in L^2. On compact subsets of O_{reg}, abelian gauge compactness and the equations give smooth subsequential limits. On each covering ball an isolated point is removable: finite-energy abelian curvature permits a $W^{1,2}$ Coulomb gauge, and the bounded spinor and the elliptic equations bootstrap the fields across the point. The extension respects the finite deck action; trivial holonomy is not assumed. The limiting real classes are the restrictions a_0. Since g was arbitrary, this supplies actual fields for every orbifold metric, including the conformal representatives used below.

On a rescaled core the same two-form satisfies $|F_A^+|_{g_i} \leq Ca^2 \to 0$. Its scale-invariant L^2 bound gives a weak limit F_i with $dF_i = 0$ and $F_i^+ = 0$. Elliptic regularity makes it an L^2 harmonic form. Pairing against compactly supported closed two-forms identifies its class as a_i. ALE L^2 Hodge theory identifies this cohomology with $\mathrm{im}(H^2_c(N_i) \to H^2(N_i))$, which here equals $H^2(N_i)$. In particular $a_i^2 \leq 0$. A finite diagonal extraction places the limits of all supplied classes on the same fixed core metrics. Their real linear combinations are also anti-self-dual, so $v_i^2 \leq 0$ for every $v \in \mathrm{span}_{\mathbb{R}}\,\mathcal{C}$. This proves (4).

For each core define

$$e_i = \frac{1}{8\pi^2}\int_{N_i}(|W^+|^2+|W^-|^2), \qquad t_i = \frac{1}{12\pi^2}\int_{N_i}(|W^+|^2-|W^-|^2).$$

Passing to the limit in the closed characteristic identities along g_a gives $\chi_{\mathrm{curv}}(O) = \chi(M) - \sum e_i$ and $\tau_{\mathrm{curv}}(O) = \tau(M) - \sum t_i$. Thus

$$T_{\mathrm{curv}}(O) = T(M) - \sum_i (2e_i + 3t_i) = T(M) - \sum_i D_i. \tag{7}$$

For any fixed g, convexity of the norm on H_g^+ implies that some supplied class has self-dual square at least B_O. The classical mixed monopole inequality, applied in invariant orbifold charts, is

$$\int_O (s - \sqrt{6}|W^+|)^2 \geq 72\pi^2 B_O.$$

Weighted Cauchy–Schwarz gives

$$\int_O \left(\frac{s^2}{24} + 2|W^+|^2\right) \geq \frac{8\pi^2}{3} B_O.$$

Finally $4\pi^2 T_{\mathrm{curv}}(O)$ equals the last integral minus $P_{\mathrm{orb}}/2$. Combining with (7) proves (5). □

The ingredients are classical curvature inequalities, abelian compactness, removability and ALE Hodge theory. The content assembled here is the full-span restriction, the preserved convex budget and the multiple-core obstruction. No classification of all Ricci-flat ALE spaces is used.

Corollary 2. *Suppose $M = X\#k\overline{\mathbb{CP}}^2$ has all supplied sign classes $K + \sum_{j=1}^k \epsilon_j E_j$, where $K^2 = A > 0$, $K \cdot E_j = 0$, $E_i \cdot E_j = -\delta_{ij}$ and $T(M) = A - k$. Set $\Delta_K = -\sum_i K_i^2 \geq 0$. Then*

$$P_{\mathrm{orb}} \geq 8\pi^2\left(k - \frac{A}{3} + \sum_i D_i + \frac{2}{3}\Delta_K\right).$$

Consequently a sequence with $\limsup P < 8\pi^2 k$ cannot have this smooth compact orbifold limit if $\sum D_i + \frac{2}{3}\Delta_K \geq A/3$.

Indeed the average sign class is K, and lower semicontinuity on compact regular subsets gives $P_{\mathrm{orb}} \leq \liminf P$. A neck energy equality is unnecessary. For r reversed Ricci-flat ASD rational balls with end groups of orders d_i^2, the criterion becomes

$$A \leq 12\sum_{i=1}^r (1 - d_i^{-2}); \qquad A \leq 9r \text{ suffices.} \tag{8}$$

3 Ordinary rational balls are actual competing topologies

Proposition 3. *Every $X\#\overline{\mathbb{CP}}^2$ contains the ordinary oriented rational ball $B_{n,1}$ for each odd $n \geq 3$. Its fundamental group maps trivially into that of the ambient manifold. It carries a Ricci-flat Kähler ALE metric in the indicated orientation with*

$$P(N) = 0, \qquad W^+ = 0, \qquad \int_N |W^-|^2 = 8\pi^2(1 - n^{-2}).$$

Proof. Reversing the orientation of a nonsingular conic in $\mathbb{CP}^2$ gives a sphere of square -4 in the $\overline{\mathbb{CP}}^2$ summand, disjoint from the connected-sum ball. Khodorovskiy's theorem embeds $B_{n,1}$ into its normal disk bundle V_{-4} for odd $n \geq 3$. This bundle is simply connected, proving the claim about fundamental groups. Suvaina's theorem supplies the Ricci-flat Kähler ALE metric with end group of order n^2. Scalar-flat Kähler geometry gives $W^+ = 0$ and ALE Gauss–Bonnet gives the displayed energy. □

This combines classical theorems; it is not a new rational-ball embedding. The smoothing (6) produces actual fixed-manifold degenerations through each such core. Both correction terms in the corollary vanish: $D_i = 0$ and $H^2(N_i;\mathbb{R}) = 0$. Thus ordinary ASD rational balls are invisible to these two improvements, whereas reversed ones satisfy (8).

4 An actual subthreshold bubbling sequence

Proposition 4 (Using the inherited strict insertion theorem)**.** *There is a numerical Godeaux surface Y with $H_1(Y;\mathbb{Z}) = \mathbb{Z}_4$ such that $M = Y\#\overline{\mathbb{CP}}^2$ admits smooth metrics g_a and a constant q_0 satisfying*

$$\frac{16\pi^2}{3} \le P_*(M) \le q_0 < 8\pi^2, \qquad P(g_a) \longrightarrow q_0,$$

while a nonflat ordinary Ricci-flat rational-ball bubble forms. The manifold M admits no Einstein metric.

Proof. Kazanova's Proposition 2.6 constructs a $\mathbb{Q}$-Gorenstein degeneration of such a smooth Y to a surface Z with ample canonical divisor and exactly one Wahl singularity $\frac{1}{4}(1,1)$. Its local smoothing is the ordinary $B_{2,1}$. Orbifold Aubin–Yau gives a negative Kähler–Einstein metric h_0 on Z. At a different, smooth point apply (3) inside a fixed ordinary chart. The negative logarithmic term gives a metric h on the one-point blow-up orbifold O with $q_0 = P_{\mathrm{orb}}(h) < 8\pi^2$. It equals h_0 near the Wahl point. The insertion is local, so its proof is unaffected by the distant quotient point.

Insert the scale-a Ricci-flat ALE smoothing of $B_{2,1}$ at that point. Its order-four expansion and the bulk normal-coordinate expansion are

$$a^2 g_N = \delta + O(a^4 r^{-4}), \qquad h = \delta + O(r^2).$$

Matching at $\rho = a^{1/2}$ gives annular curvature $O(1)$ and annular energy $o(1)$. The exact inner region has $P = 0$; the removed bulk ball also has $P = 0$. Hence $P(g_a) \to q_0$. The smoothing topology and the blow-up away from the singular point give the same fixed smooth manifold M for all small $a > 0$. After rescaling, the inserted region exhausts the nonflat ALE metric, so curvature is unbounded in the fixed bulk scale.

Here $K_Y^2 = A = 1$ and $T(M) = 0$. The classical all-metric monopole bound is $P(g) \ge 8\pi^2(\frac{2}{3}A - T) = 16\pi^2/3$, excluding Einstein metrics. This bound and the constructed sequence prove the claimed interval. □

The example is a torsion-$\mathbb{Z}_4$ Godeaux surface, not the classical torsion-$\mathbb{Z}_5$ surface. Kazanova's Corollary 2.5 excludes this particular ample-canonical single-Wahl degeneration for the latter family. That algebraic exclusion alone says nothing about all smooth embeddings or all Riemannian degenerations on it.

5 The remaining comparison is variational

The preceding proposition shows that the sublevel $P < 8\pi^2$ need not be smoothly compact even on a fixed non-Einstein example. It does not say that $q_0 = P_*(M)$, so it is neither a nonattainment theorem nor a solution of Q3.

For a specified orbifold boundary obtained by these Ricci-flat smoothings, (6) gives

$$P_*(M) \le P_{*,\mathrm{orb}}(O).$$

If a genuine minimizing sequence has this compact orbifold limit, lower semicontinuity gives the reverse comparison through its limit metric. Thus that branch requires equality with an optimized

singular problem. To exclude it one needs a strict variational comparison, or information about genuine minimizing sequences that is absent from an arbitrary sublevel bound. Partial collapse remains a separate possible branch.

The multiple-core theorem strengthens the range already excluded. The subthreshold construction proves that the complementary orientation is a real possibility at the relevant energy scale. These are complementary conclusions about the same attainment problem, not two new research topics.

References

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